Department of Mathematics and Statistics, School of Arts and Science, Qatar University.

Unit Name:	Mathematics for EE	Instructor:	Dr. Mohamed Mabrok
Section:	L01	Location:	Lecture Room
Exam :	**Fall 2024, Final Exam**	Time and Date:	December 11, 2024

Student Name:	
Student ID:	

- No credit will be given to answers for non-requested questions.
- Only non-plotting calculators are permitted. The use of the cell phones is absolutely prohibited.
- Shut off your cell phone completely.
- Make sure that you are feeling comfortable before entering the class for the test.
- Full mark will be given for complete answers for all questions.
- Cheating is an academic violation according to Qatar University rules and regulations, and in some cases, it may result in final dismissal from the University.
- Students should not under any circumstances commit or participate in any cheating attempt or any act that violates student code of conduct.

Question Number	Student Marks	Maximum Marks
1		**7**
2		**6**
3		**15**
4		**12**
5		**15**
6		**15**
Total / (Max)		**70**

1. Solve the following differential equation:

$$(x^2 + \ln(y))dx + \left(\frac{x}{y} + 2\right) dy = 0.$$

6 Marks

2. Find the general solution of the following differential equation:

$$xy'' - y' = 4x^2 e^x$$

6 Marks

3. Compute the following:

$$A)\ \mathscr{L}^{-1}\left(\frac{3s+8}{s^2+2s+5}\right)$$

$$B)\ \mathscr{L}\left(f(t)\right), \quad \text{where} \quad f(t)=\begin{cases} t & \text{if } t<6 \\ -8+(t-6)^2 & \text{if } t\geq 6 \end{cases}$$

$$C)\ \mathscr{L}^{-1}\left(\frac{5\left(se^{-3s}-e^{-4s}\right)}{s^2+4}\right)$$

15 Marks

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4. Use Laplace transforms, find the general formula for the solution of the initial value problem for any $f(t)$.

$$y'' + 3y' = f(t), \quad y(0) = 0, \quad y'(0) = 0$$

Then, use the obtained general form and

$$f(t) = t$$

to find the solution (*You have to compute the integration*). 12 Marks

5. Convert the following differential equation into a system of first-order DE's, then, solve the system using matrices.

$$y''' + 3y'' + 2y' = 0$$

15 Marks

6. Solve the following:

(a) Find the real and imaginary part of the following:

$$\frac{(2-i)^2-(1+3i)^2}{i+2}$$

(b) Suppose that:

$$z_1 = 1+2i, \quad z_2 = -3-3i, \quad z_3 = 2i+4.$$

-Compute $z_1\,\bar{z_3}$.
- Write z_1 the polar form.

(c) Find the all roots of $w^3 = 2+i$

(d) Show that $f(z) = 2\bar{z}$ is not differentible

15 Marks

Algebra

- $\sin\theta = \frac{1}{\csc\theta}$
- $\csc\theta = \frac{1}{\sin\theta}$
- $\cos\theta = \frac{1}{\sec\theta}$
- $\sec\theta = \frac{1}{\cos\theta}$
- $\tan\theta = \frac{1}{\cot\theta}$
- $\cot\theta = \frac{1}{\tan\theta}$
- $\sin^2\theta + \cos^2\theta = 1$
- $1 + \tan^2\theta = \sec^2\theta$
- $1 + \cot^2\theta = \csc^2\theta$
- $\cos^2(x) = \frac{1}{2}(1 + \cos(2x))$
- $\sin^2(x) = \frac{1}{2}(1 - \cos(2x))$
- $\cos^2(x) = 1 - \sin^2(x)$
- $\sin^2(x) = 1 - \cos^2(x)$

Differentiation

- $\frac{\mathrm{d}}{\mathrm{d}x} c = 0$
- $\frac{\mathrm{d}}{\mathrm{d}x} x^n = n x^{n-1}$
- $\frac{\mathrm{d}}{\mathrm{d}x} (ax \pm b)^n = a\, n\, (ax \pm b)^{n-1}$
- $\frac{\mathrm{d}}{\mathrm{d}x} \ln|x| = \frac{1}{x}$
- $\frac{\mathrm{d}}{\mathrm{d}x} \ln|ax \pm b| = \frac{a}{ax \pm b}$
- $\frac{\mathrm{d}}{\mathrm{d}x} e^{ax} = a\, e^{ax}$
- $\frac{\mathrm{d}}{\mathrm{d}x} \sin(ax) = a\, \cos(ax)$
- $\frac{\mathrm{d}}{\mathrm{d}x} \cos(ax) = -a\, \sin(ax)$
- $\frac{\mathrm{d}}{\mathrm{d}x} f(x)\, g(x) = f(x) \cdot \frac{\mathrm{d}g(x)}{\mathrm{d}x} + g(x) \cdot \frac{\mathrm{d}f(x)}{\mathrm{d}x}$ *Product rule.*
- $\frac{\mathrm{d}}{\mathrm{d}x} \frac{f(x)}{g(x)} = \frac{g(x) \cdot \frac{\mathrm{d}f(x)}{\mathrm{d}x} - f(x) \cdot \frac{\mathrm{d}g(x)}{\mathrm{d}x}}{(g(x))^2}$ *Quotient rule.*
- $\frac{\mathrm{d}}{\mathrm{d}x} f(g(x)) = f'(g(x)) \cdot g'(x)$ *Chain rule.*

Integration

- $\int a\, dx = ax + c$
- $\int x^n\, dx = \frac{x^{n+1}}{n+1} + c \qquad n \neq -1$
- $\int f(x)^n f'(x)\, dx = \frac{f(x)^{n+1}}{n+1} + c \qquad n \neq -1$
- $\int \frac{a}{x}\, dx = a\, \ln|x| + c$
- $\int \frac{1}{ax+b}\, dx = \frac{1}{a} \ln|ax+b| + c$
- $\int e^{ax}\, dx = \frac{1}{a}\, e^{ax} + c$
- $\int \frac{f'(x)}{f(x)}\, dx = \ln|f(x)| + c$
- $\int \sin(ax)\, dx = \frac{-1}{a}\, \cos(ax) + c$
- $\int \cos(ax)\, dx = \frac{1}{a}\, \sin(ax) + c$
- $\int \sec^2(ax)\, dx = \frac{1}{a}\, \tan(ax) + c$
- $\int \csc^2(ax)\, dx = -\frac{1}{a}\, \cot(ax) + c$
- $\int \tan(ax)\, dx = -\frac{1}{a}\, \ln(\cos(ax)) + c$
- $\int \cot(ax)\, dx = \frac{1}{a}\, \ln(\sin(ax)) + c$
- $\int \sec(ax)\tan(ax)\, dx = \frac{1}{a}\, \sec(ax) + c$
- $\int \csc(ax)\cot(ax)\, dx = -\frac{1}{a}\, \csc(ax) + c$
- $\int \sec(ax) dx = \frac{1}{a}\, \ln(\sec(ax) + \tan(ax)) + c$
- $\int \csc(ax)\, dx = \frac{-1}{a}\, \ln(\csc(ax) - \cot(ax)) + c$
- $\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1}\left(\frac{x}{a}\right) + C$
- $\int \frac{1}{x^2 + a^2} dx = \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right) + C$
- $\int \frac{1}{x\sqrt{x^2 - a^2}} dx = \frac{1}{a} \sec^{-1}\left(\frac{x}{a}\right) + C$

Table of Laplace Transforms

	$f(t)=\mathfrak{L}^{-1}\{F(s)\}$	$F(s)=\mathfrak{L}\{f(t)\}$		$f(t)=\mathfrak{L}^{-1}\{F(s)\}$	$F(s)=\mathfrak{L}\{f(t)\}$
1.	1	$\frac{1}{s}$	2.	$\mathbf{e}^{at}$	$\frac{1}{s-a}$
3.	$t^n,\quad n=1,2,3,\ldots$	$\frac{n!}{s^{n+1}}$	4.	$t^p,\ p>-1$	$\frac{\Gamma(p+1)}{s^{p+1}}$
5.	$\sqrt{t}$	$\frac{\sqrt{\pi}}{2s^{\frac{3}{2}}}$	6.	$t^{n-\frac{1}{2}},\quad n=1,2,3,\ldots$	$\frac{1\cdot 3\cdot 5\cdots(2n-1)\sqrt{\pi}}{2^n s^{n+\frac{1}{2}}}$
7.	$\sin(at)$	$\frac{a}{s^2+a^2}$	8.	$\cos(at)$	$\frac{s}{s^2+a^2}$
9.	$t\sin(at)$	$\frac{2as}{(s^2+a^2)^2}$	10.	$t\cos(at)$	$\frac{s^2-a^2}{(s^2+a^2)^2}$
11.	$\sin(at)-at\cos(at)$	$\frac{2a^3}{(s^2+a^2)^2}$	12.	$\sin(at)+at\cos(at)$	$\frac{2as^2}{(s^2+a^2)^2}$
13.	$\cos(at)-at\sin(at)$	$\frac{s(s^2-a^2)}{(s^2+a^2)^2}$	14.	$\cos(at)+at\sin(at)$	$\frac{s(s^2+3a^2)}{(s^2+a^2)^2}$
15.	$\sin(at+b)$	$\frac{s\sin(b)+a\cos(b)}{s^2+a^2}$	16.	$\cos(at+b)$	$\frac{s\cos(b)-a\sin(b)}{s^2+a^2}$
17.	$\sinh(at)$	$\frac{a}{s^2-a^2}$	18.	$\cosh(at)$	$\frac{s}{s^2-a^2}$
19.	$\mathbf{e}^{at}\sin(bt)$	$\frac{b}{(s-a)^2+b^2}$	20.	$\mathbf{e}^{at}\cos(bt)$	$\frac{s-a}{(s-a)^2+b^2}$
21.	$\mathbf{e}^{at}\sinh(bt)$	$\frac{b}{(s-a)^2-b^2}$	22.	$\mathbf{e}^{at}\cosh(bt)$	$\frac{s-a}{(s-a)^2-b^2}$
23.	$t^n\mathbf{e}^{at},\quad n=1,2,3,\ldots$	$\frac{n!}{(s-a)^{n+1}}$	24.	$f(ct)$	$\frac{1}{c}F\left(\frac{s}{c}\right)$
25.	$u_c(t)=u(t-c)$ Heaviside Function	$\frac{\mathbf{e}^{-cs}}{s}$	26.	$\delta(t-c)$ Dirac Delta Function	$\mathbf{e}^{-cs}$
27.	$u_c(t)f(t-c)$	$\mathbf{e}^{-cs}F(s)$	28.	$u_c(t)g(t)$	$\mathbf{e}^{-cs}\mathfrak{L}\{g(t+c)\}$
29.	$\mathbf{e}^{ct}f(t)$	$F(s-c)$	30.	$t^n f(t),\quad n=1,2,3,\ldots$	$(-1)^n F^{(n)}(s)$
31.	$\frac{1}{t}f(t)$	$\int_s^\infty F(u)\,du$	32.	$\int_0^t f(v)\,dv$	$\frac{F(s)}{s}$
33.	$\int_0^t f(t-\tau)g(\tau)\,d\tau$	$F(s)G(s)$	34.	$f(t+T)=f(t)$	$\frac{\int_0^T \mathbf{e}^{-st}f(t)\,dt}{1-\mathbf{e}^{-sT}}$
35.	$f'(t)$	$sF(s)-f(0)$	36.	$f''(t)$	$s^2F(s)-sf(0)-f'(0)$
37.	$f^{(n)}(t)$	$s^nF(s)-s^{n-1}f(0)-s^{n-2}f'(0)\cdots-sf^{(n-2)}(0)-f^{(n-1)}(0)$			